```
B·u + h*v·u = D·u + g*w·u → B·u = D·u + g*w·u
```

And as such we eliminate the unknown h and solve the equation for g:

$$g = \frac{(B-D)\cdot\vec{u}}{\vec{w}\cdot\vec{u}} = \frac{(B_x-D_x, B_y-D_y)\cdot(A_y-B_y, B_x-A_x)}{(D_x-C_x, D_y-C_y)\cdot(A_y-B_y, B_x-A_x)} =$$
$$= \frac{B_xA_y - B_xB_y - D_xA_y + D_xB_y + B_yB_x - B_yA_x - D_yB_x + D_yA_x}{D_xA_y - D_xB_y - C_xA_y + C_xB_y + D_yB_x - D_yA_x - C_yB_x + C_yA_x} =$$
$$= \frac{D_x(B_y-A_y) + D_y(A_x-B_x)}{(D_x-C_x)(A_y-B_y) + (D_y-C_y)(B_x-A_x)}$$

Similarly, we can define vector z = $(-w_y, w_x)$, such that $z \cdot w = 0$, and derive a formula for h:

$$h = \frac{(D-B)\cdot\vec{z}}{\vec{v}\cdot\vec{z}} = \frac{B_x(D_y-C_y) + B_y(C_x-D_x)}{(B_x-A_x)(C_y-D_y) + (B_y-A_y)(D_x-C_x)}$$

All it's left is reasoning about the meaning of these two solutions; looking at figure 15.19, can you tell which one, between h and g is, for this example, larger than 1?

If you notice, in this particular example vector BP is smaller than vector BA, and as such h must have a value between 0 and 1; the scalar g, instead must be larger than 1, because vector DP is longer than DC.

For the former situation, clearly P is within the segment BA, while for the latter, it's clearly outside DC.

Listing 15.8 Method vectorScalingFactor